اللهم باسمك نبدأ، وبك نستعين

Genetics test bank

Blueprint of Life.

﴿وَلَقَدْ خَلَقْنَا الإِنسَانَ مِن سُلالَةٍ مِّن طِينٍ ثُمَّ جَعَلْنَاهُ نُطْفَةً فِي قَرَارٍ مَّكِينٍ﴾

(سورة المؤمنون: 12)

First mid

Med25

1. What is an alternative form of a gene called?

A) Chromosome
B) Genome
C) Allele
D) Locus

2. Parents are phenotypically normal but have two children with an autosomal dominant disease. What is the most likely explanation?

A) Gonadal mosaicism
B) Somatic mosaicism
C) Complete penetrance
D) Anticipation

3. Consanguinity increases the risk of which type of genetic disorder?

A) Autosomal dominant disorders
B) X-linked dominant disorders
C) Autosomal recessive disorders
D) Mitochondrial disorders

4. Which of the following is an example of an X-linked recessive disorder?

A) Marfan syndrome
B) Huntington disease
C) Red-green color blindness
D) Cystic fibrosis

1	2	3	4
C	A	C	C

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Med25

1. Which of the following diseases is an example of pleiotropy?

A) Down syndrome
B) Marfan syndrome
C) Sickle cell anemia
D) Turner syndrome

2. What is the definition of a gene?

A) A sequence of RNA that forms proteins
B) A unit of energy in the cell
C) A hereditary unit of DNA that encodes a functional product
D) A structure found in the cytoplasm

3. What is meant by a gene variant (mutation)?

A) Movement of a gene within the genome
B) A reversible change in a protein structure
C) A permanent change in the DNA sequence
D) Temporary modification of gene expression

4. What is the DNA sequence repeat associated with Huntington disease?

A) CGG
B) GAA
C) CTG
D) CAG

1	2	3	4
B	C	C	D

Med25

1. Which of the following best describes autosomal dominant inheritance in a pedigree?

A) Skips generations
B) Affects only males
C) Appears in every generation (vertical transmission)
D) Affects only females

2. What is the definition of translocation?

A) Deletion of a chromosome segment
B) Duplication of genetic material
C) Exchange of genetic material between non-homologous chromosomes
D) Inversion of a chromosome segment

3. A child presents with muscle weakness, neurological symptoms, and a history of similar symptoms affecting multiple siblings from the same mother. The father is unaffected. Which type of inheritance best explains this pattern?

A) Autosomal dominant inheritance
B) Autosomal recessive inheritance
C) X-linked inheritance
D) Mitochondrial inheritance

4.What are the main structural components of a gene?

A) Coding regions that are directly translated into proteins along with RNA processing elements
B) Non-coding DNA sequences involved in regulation and control of gene expression
C) Exons, introns, and regulatory sequences such as promoters
D) Cellular structures involved in protein synthesis such as ribosomes and transfer RNA

1	2	3	4
C	C	D	C

Med25

1. Where does protein synthesis occur in the cell?

A) Nucleus
B) Ribosome
C) Golgi apparatus
D) Lysosomes

2. Which of the following terms describes the inheritance of two different homologous chromosomes from the same parent?

A) Uniparental isodisomy
B) Chromosomal deletion
C) Nondisjunction
D) Uniparental heterodisomy

3. Which condition affects both males and females equally and is an example of pleiotropy?

A) Hemophilia
B) Turner syndrome
C) Marfan syndrome
D) Color blindness

4. During which process is mRNA produced?

A) Translation
B) Replication
C) Post-translational modification
D) Transcription

1	2	3	4
B	D	C	D

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Med25

1. What is the number of autosomes in a male gamete?

A) 44
B) 23
C) 22
D) 46

2. Which chromosome is considered a satellite chromosome?

A) Chromosome 1
B) Chromosome 6
C) Chromosome 11
D) Chromosome 14

3. What is the normal karyotype of a healthy female?

A) 45,X
B) 47,XXX
C) 46,XY
D) 46,XX

4. Which of the following chromosomes is classified as metacentric?

A) Chromosome 13
B) Chromosome 14
C) Chromosome 3
D) Chromosome 21

1	2	3	4
C	D	D	C

Med25

1. What is the primary function of the centromere?

A) To initiate DNA replication during S phase
B) To regulate gene transcription along the chromosome
C) To allow attachment of spindle microtubules during cell division
D) To maintain chromosome length during cell division

2. What does the cytogenetic notation 7q15 indicate?

A) Chromosome 7 short arm, first region, fifth band
B) Chromosome 7 long arm, fifth region, first band
C) Chromosome 7 short arm, fifth region, first band
D) Chromosome 7 long arm, first region, fifth band

3. Loss of the paternal chromosome segment 15q11–q13 leads to which condition?

A) Angelman syndrome
B) Down syndrome
C) Turner syndrome
D) Prader-Willi syndrome

4. Which enzyme is responsible for unwinding DNA during replication?

A) DNA polymerase
B) RNA polymerase
C) Ligase
D) Helicase

1	2	3	4
C	D	D	D

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Med25

1. The severity of a genetic disease varies among individuals within the same family. What is the most likely explanation?

A) Complete penetrance
B) Genetic anticipation
C) Variable expressivity
D) Pleiotropy

2. A child is diagnosed with Angelman syndrome. Which gene is most likely affected?

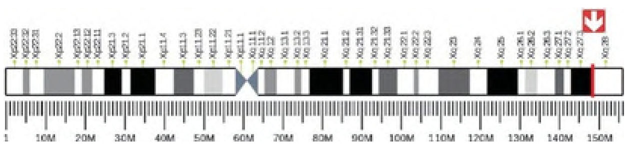
A) FMR1
B) DMD
C) CFTR
D) UBE3A

3. From which cell are mitochondrial diseases transmitted to offspring?

A) Sperm cell
B) Somatic cell
C) Stem cell
D) Oocyte

4.A chromosome ideogram is shown with labeled bands. Which of the following regions is closest to the telomere?

A) Xq12
B) Xq21
C) Xq26
D) Xq28



1	2	3	4
C	D	D	D

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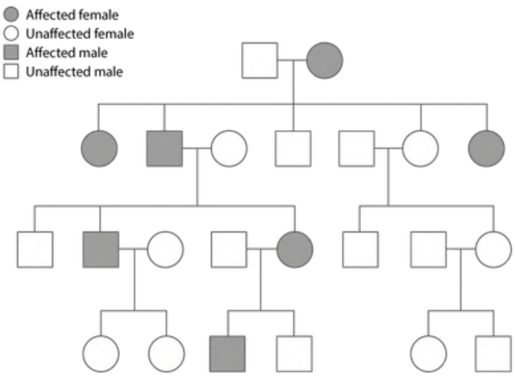
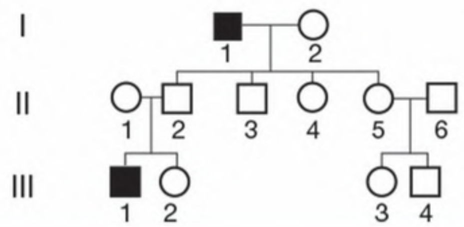
1.What does this pedigree most likely demonstrate?

A) Autosomal recessive inheritance
B) X-linked recessive inheritance
C) X-linked dominant
D) Autosomal dominant inheritance with reduced penetrance



2.What does this pedigree most likely demonstrate?

A) Autosomal dominant inheritance
B) Autosomal recessive inheritance
C) X-linked recessive inheritance
D) Mitochondrial inheritance



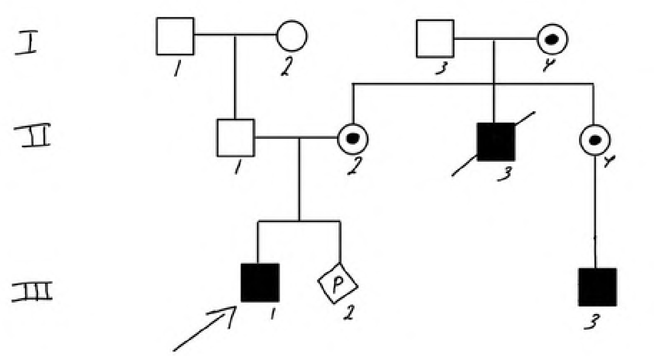
1	2	3	4
D	C		

Med25

Adel, a 10-year-old boy, presented with dysmorphic facies and mental retardation. He has a similarly affected maternal uncle (deceased). His mother's sister also has a similarly affected son. In addition, the maternal grandmother is a carrier of Duchenne muscular dystrophy. The mother is currently pregnant. The family came to the genetic clinic seeking genetic counseling.

Tasks:

1. Draw a pedigree for this family.



2. What is the most likely mode of inheritance?

X-linked recessive

3. What is the risk to the mother's fetus?

- 50% carrier female
- 50% unaffected female
- 50% affected male
- 50% unaffected male

Questions

Year : MED24

1. Which chromosome has an appendage-like structure (satellite)?

A) Chromosome 10
B) Chromosome 16
C) Chromosome 14
D) Chromosome 1

2. When breaks occur on each arm of a chromosome and the segment between the breaks is reconstituted, which type of chromosome rearrangement is this?

A) Ring chromosome
B) Pericentric inversion
C) Paracentric inversion
D) Translocation

3. Which technique would be used in recurrent miscarriages?

A) Karyotyping
B) FISH
C) Cytogenetic analysis
D) Chromosomal microarray

4. Which of the following is most commonly detected by cytogenetic testing?

A) Balanced translocation
B) Triple repeat
C) Microdeletions
D) Point mutations

1	2	3	4
C	B	A	A

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Questions

Year : MED24

1. In FISH testing, how is a microdeletion detected?

A) By specific chromosome stain
B) By detecting point mutations
C) By analyzing banding patterns
D) By detecting trisomies

2. What is the probe of FISH used for microdeletion and duplication?

A- Specific locus
B- Centromeric
C- Telomere
D- Whole painting

3. What enzyme is used to unwind the double helix of DNA?

A-ligase
B-DNA polymerase
C-helicase
D-topoisomerase

4. The most common numerical cause of Turner syndrome is?

A)46,X,r(X)
B)46,X,i (Xq)
C)46,X,del(X) (q12)
D)45,X

1	2	3	4
A	A	C	D

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Questions

Year : MED24

1. A female came to the hospital, and she was pregnant. The doctor told her that her son was suffering from unbalanced rearrangement. What could it be?

A-Robertsonian translocation
B-Derivative chromosome
C-Deletion

2. What is a characteristic of X-linked dominant?

A- Affects males and females equally but more in females
B- More severe in females
C- Only daughters of the affected father will be affected
D- Male-to-male transmission

3. Genomic imprinting is an important factor in gene variation. Which of the following is an example of genital imprinting?

A- Angelman syndrome and Prader-Willi syndrome
B- Angelman syndrome and Cystic fibrosis
C- Beckwith-Wiedemann syndrome and Silver-Russell syndrome
D- Huntington's disease and Duchenne muscular dystrophy

4. How can Prader-Willi syndrome (PWS) occur?

A. Uniparental disomy of the maternal chromosome
B. Uniparental disomy of the paternal chromosomes
C. Deletion of the paternal allele in the 15q11-q13 region
D. Mutation of the UBE3A gene on the maternal chromosome

1	2	3	4
C	C	A	A & C

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Questions

Year : MED24

1. What is a compound heterozygous mutation?

A- When a child inherits an autosomal recessive disorder from both parents, who have different mutations on some gene
B- When both alleles of a gene carry the some mutation
C- When a person has one mutated allele and one normal allele in a dominant disorder
D- When two different genes are mutated, leading to the same disease

2. Which cell can't be used for chromosomal analysis?

A- Skin cells
B- WBCs
C- Hepatocyte
D- RBCS

3. What is the most common cause of triploidy?

A) Fertilization of an egg by two sperm
B) Failure of meiotic division leading to a diploid egg
C) Duplication of the paternal genome after fertilization
D) Nondisjunction during maternal meiosis

4. A couple visits a genetic lab for testing after their baby is born with dysmorphic features. The baby is diagnosed with an unbalanced chromosomal abnormality. Which of the following karyotypes is most consistent with this condition?

A) 46, XX
B) 45, XX, der(21:21) (410,q10)
C) 46, XX, del(4) (q10)
D) 45, XX, der(14,21) (410,910)

1	2	3	4
A	D	A	C

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Questions

Year : MED24

1. What is the most common cause of Turner syndrome?

A) Monosomy X
B) Deletion of part of the X chromosome
C) Uniparental disomy of the X chromosome
D) Balanced translocation involving the X chromosome

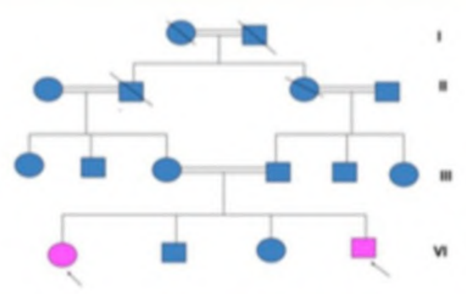
2. Which syndrome does the image represent?

A. Patau syndrome
B. Cri-du-chat syndrome
C. Wolf-Hirschhorn syndrome
D. Down syndrome

3. The correct karyotype is:

A. 46, X, r(X)
B. 47, Y, r(X)
C. 46, XX, r(X)
D.47, X, r(X)

4. This pedigree shows:



A. Only females are affected
B. Gene skip generation
C. Gene without skip generation
D. One parent must be affected

1	2	3	4
A	B	A	B

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Questions

Year : MED24

1. In autosomal recessive inheritance, what is the chance of being heterozygous unaffected (carrier)?

A-25%
B-50%
C-75%
D-100%

2. This pedigree shows:

A) Polyploidy
B) Heteroploidy
C) Somatic mosaicism
D) Autosomal recessive inheritance

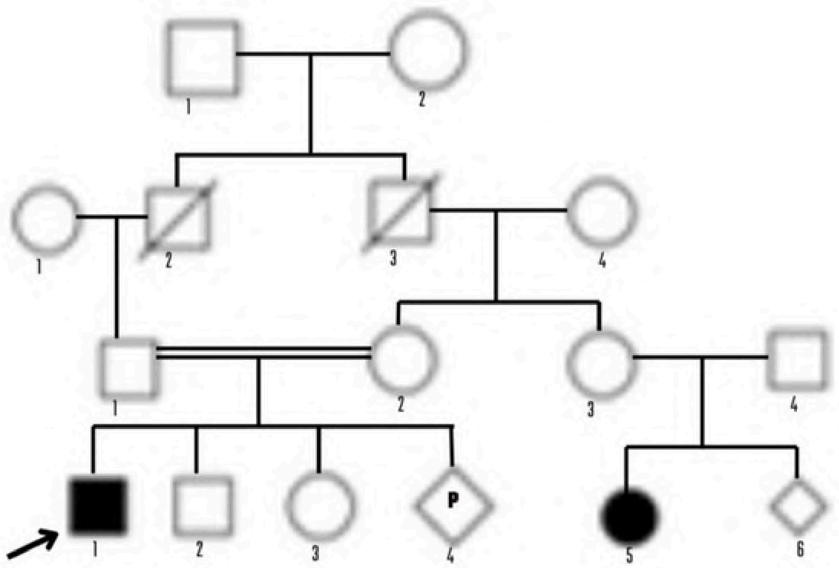
1	2	3	4
B	B		

Short essay Q

Year : MED24

Draw a three-generation pedigree for this family

A 3-year-old boy diagnosed with cystic fibrosis was referred to the genetics clinic. He has one healthy brother and one healthy sister. His mother is currently pregnant, The boy's parents are first cousins, as their fathers are brothers. All grandparents are deceased,On the maternal side, the mother has a sister who has a daughter affected with cystic fibrosis. This affected daughter later experienced a miscarriage.



1	2	3	4

Genetics Final Season (The 100)

1)Terminology that describe an Alternative form of gene found on the same locus on a pair of homologous chromosome :
A. Allele
B. Gene
C. Heterozygosity
D. Homozygosity

2)what is the characteristic of balanced translocation?
A. no change in gene dose
B. No DNA mutation
C. Severe mutation in the DNA
D. Chromosomal aberration with congenital anomalies

3) which of the following is true about mitosis
A. The daughter cell haploid
B. Division of centromere and chromatids will separate during anaphase
C. Two sister chromatid goes in same side
D. The chromosome pair will separate in anaphase

4) What is the term used to describe a double break in the short arm of a chromosome and replaced in the region between the breaks inverted?
A. Pericentric inversion
B. Isochromosome
C. Paracentric inversion
D. Insertion

1	2	3	4
A	A	B	C

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Med 22

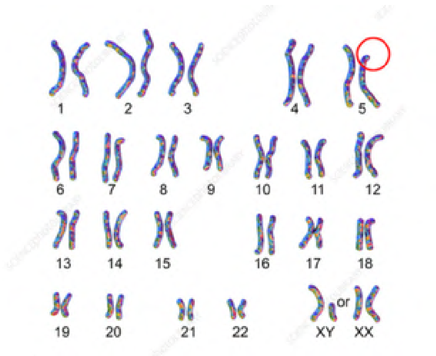
Genetics Final Season (The 100)

1) which of the following represents delation of short in arm of chromosome 10 in male

A. 46,XX,del(10)(p11.2p23)
B. 46,XY,del(10)(p11.2p23)
C. 45,XY,del(10)(q11.2q23)
D. 46,XY,del(10)(p11q23)

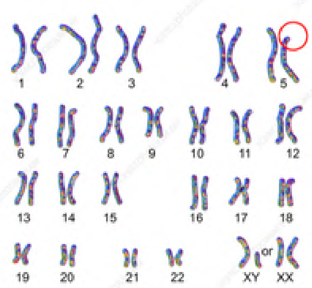
2) The karyogram below shows

A. it involve 2 breaks 5p deletion
B. it involve 2 breaks 5p that lead to inversion
C. it involve single break of 5p in the terminal part and leads to chromosomal loss
D. it involve single break of 5q in the terminal part and leads to chromosomal loss



3) what is the condition

A. Edward syndrome
B. William syndrome
C. Cri-du-chat syndrome



4)Which of the following is a suitable nomenclature for the karyotype below:

A. 46,XX,der(14;21)(q10,q10)
B. 47,XX,rob(14;21)(q10,q10)
C. 46,XX,rob(14;21)
D. 45,XX,rob(14;21)(q10,q10)

1	2	3	4
B	C	C	D

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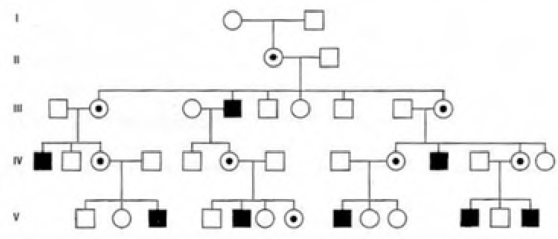
Genetics Final Season (The 100)

1) The pedigree is in consistence with one of the following disorders:

A. Rickets
B. A-thalassemia
C. G6PD deficiency
D. Hurler syndrome



2) A male came with an intellectual disability, long face, protruded ears. The genetic testing showed mutation in FMR1. What should we ask for to see if this is an inherited or new mutation?

A. Genetic testing for the mother
B. Genetic testing for the father
C. Genetic testing for both parents
D. Genetic testing for the siblings

3) A male came with an intellectual disability, long face, protruded ears. The genetic testing showed mutation in FMR1. What is the mode of inheritance?

A. X-linked
B. Autosomal recessive
C. Autosomal dominant

4) A 30-year-old female has ovarian insufficiency. The son has (ear problems). Her 65 year-old father got a tremor. What describes their conditions?

A. the father is a carrier for pre-mutation 80 repeats of CGG , the son has a mutation of more than 200 repeats
B. Both son and father have a mutation in FMR1 gene

1	2	3	4
C	A	A	A

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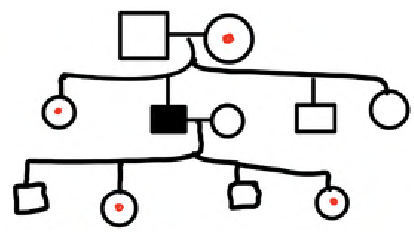
Genetics Final
Season (The 100)

1) 46,XY, t(7;11)
A. Male with reciprocal translocation between chromosome 7 and chromosome 11
B. Male with insertion between chromosome 7 and chromosome 11

2) carrier female with rob(14;21), what is the recurrence risk for getting baby with down syndrome
A. 10-15%
B. 1-3%

3) A 6-year-old male patient with seizure, intellectual disability, inappropriate laughter, microocephaly, unsteady gait, what is the most likely genetic mechanism?
A. Loss of maternal contribution of region on chromosome 15
B. Loss of paternal contribution of region on chromosome 15
C. Deletion of short arm chromosome 5 from father
D. Deletion of short arm chromosome 5 from mother

4) what is the mode of inheritance from this pedigree
A. Autosomal dominant
B. Autosomal recessive
C. X linked dominant
D. X linked recessive



1	2	3	4
A	A	A	D

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Genetics Final Season (The 100)

1) The normal individual will have only one transcriptionally active copy of the gene. What is the most common explanation of this process of gene silencing?
A. Uniparental disomy
B. Mosaicism
C. Genomic imprinting
D. New mutation

2) The following chart shows the stature of a 13 years old girl with primary amenorrhea, congenital heart disease and webbed neck. What is the diagnosis?
A. Digeorge syndrome
B. Edward syndrome
C. Turner Syndrome
D. Marfan syndrome

3) Which of the following samples is suitable for karyotype analysis?
A. Peripheral blood
B. urine
C. saliva
D. feces

4) a child present with Microcephaly, hypotonia, and cat-like cry characteristic what would the chromosomal analysis show:
A. 5p deletion
B. 15p deletion
C. 5q deletion

1	2	3	4
C	C	A	A

Genetics Final Season (The 100)

1) Majed, 30 years old, went to the genetic clinic, married for three years, and sterile due to azoospermia, the family history was negative, and his characteristics were tall and his arms were long. What is the diagnosis?

A. Turner syndrome
B. Marfan syndrome
C. Fragile X
D. Cystic fibrosis

2) Which of the following are offspring of a consanguineous marriage at increased risk of?

A. Chromosome abnormalities
B. X-linked recessive
C. Autosomal dominant
D. Congenital malformation

3) What is a major anomaly of Marfan syndrome?

A. Arachondactylyl
B. Aortic aneurysm
C. Pes planus
D. Short stature

1	2	3	
B	D	B	

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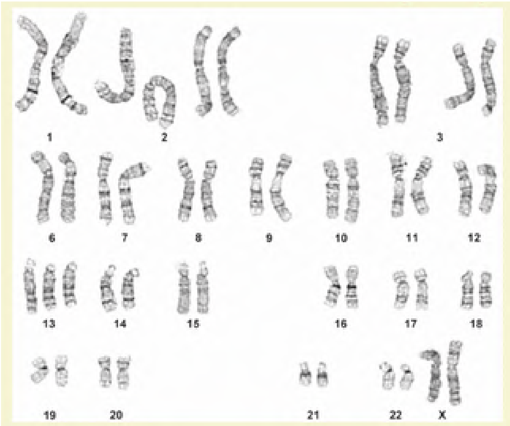
Med 22

Genetics Final
Season (The 100)

1) What does the following karyotype show?

A. Triploidy
B. Aneuploidy
C. Polyploidy



2) A female presented in the NICU with heart and renal abnormalities. After examination: microcephaly microphthalmia, hypertelorism, postaxial polydactyly. What is the provisional diagnosis?

A. patau syndrome
B. down syndrome

3) Matrilineal inheritance applies to disorders that are directly due to mutations in which of the following?

A. Alpha globin gene
B. Mitochondrial DNA
C. tumor suppressor gene
D. CYP2C9

1	2	3	
B	A	B	

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MCQ

Year : MED21

1.Which of the following genetic mechanisms is known to cause Prader–Willi syndrome?

A) Maternal chromosome 15 deletion
B) Paternal chromosome 15 deletion
C) Trisomy of chromosome 15
D) Balanced translocation involving chromosome 15

2.Which is the sequence of poly a tail that attached to mRNA?
A) AAUAAA
B) AATAAA
C) AACAAA
D) AAGAAA

3. What is the terminology used to describe the total genetic heritage material of a cell or organism?

a)Chromosome
b)Genome
c)Gene
d) Locus

4. The unit that contains nucleic acid and protein and is responsible for heredity is called?

A) chromosome
B) Genome
C) RNA
D) protein

1	2	3	4
B	A	B	A

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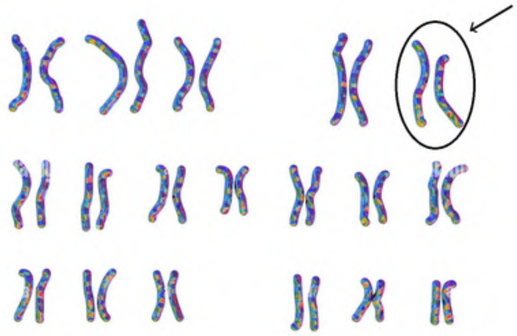
الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ

Year : MED21

1.In the following diagram, what chromosomal abnormality is shown?

A) Duplication
B) Inversion
C) Translocation
D) Deletion



2.Regarding the cell cycle the resting phase is?

A) Prophase
B) Interphase
C) Metaphase
D) Anaphase

3.Which of the following conditions is characterized by the presence of two or more different chromosomal cell lines within the same individual?

A) Mosaicism
B) Triploidy
C) Aneuploidy
D) Tetraploidy

4.A 10 year old boy came to the clinic with an autosmal dominant disorder with no family history. What is the explanation of having autosomal dominant with no family history?

A) New mutation
B) Autosomal recessive inheritance
C) X-linked inheritance
D) Uniparental disomy

1	2	3	4
D	B	A	A

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MCQ

Year : MED21

1.the mRNA sequence TAG TAG TAG if we add cytosine at the beginning what will be the new sequence?

A) CTA GTA GTAG
B) CTAGTAGTAG
C) CTA CTA GTAG
D) non sense

2.What is responsible for the integrity of the chromosome during the cell cycle and it's loss would results in the formation of a ring chromosome? ?

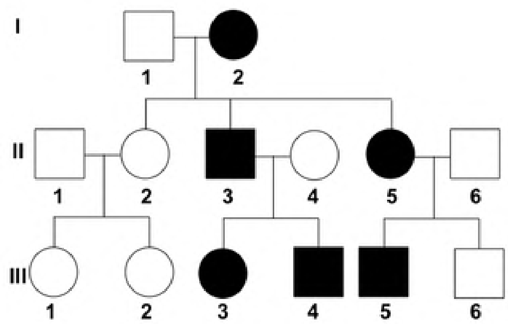
A) Histone
B) Proteins
C) Telomeres
D) Centromere

3.A 10-year-old boy came to the genetic clinic complaining of a genetic disease. Family history revealed a similarly affected sister. His father and paternal grandmother have moderate manifestations. Why did his father and paternal grandmother develop moderate manifestations?

A) Genomic imprinting
B) Variable expression
C) Somatic mosaicism
D) Polyploidy

4. What is the type of inheritance?

A) Holandric
B) mitochondrial
C) X-linked dominant
D) Autosomal dominant



1	2	3	4
A	C	B	D

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

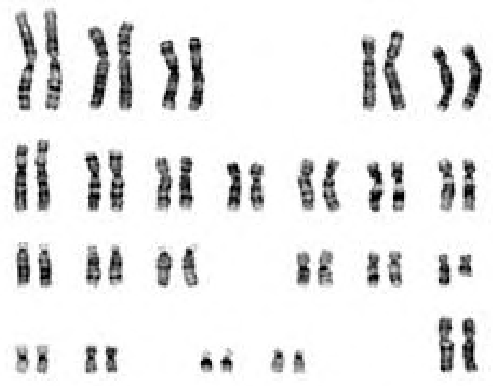
MCQ

Year : MED21

1.What is the risk of recurrence per ecah pregnancy in X-linked dominant inheritance?

A) 10%
B) 25%
C) 50%
D) 100%

2. What is demonstrated in the image below?



A) Deletion
B) Insertion
C) Isochromosome
D) Inversion

3.Which of the following is an X-linked recessive disorder?

A) Duchenne Muscular Dystrophy
B) Cystic fibrosis
C) Albinism
D) Sickle cell anemia

4. What type of disorder is Hypophosphatemic Rickets (Vitamin D resistance rickets)?

A) Autosomal dominant
B) Holandric
C) X-linked dominant
D) X-linked recessive

1	2	3	4
C	A	A	C

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ

Year : MED21

1.Which one of these causes triploidy?

A) fertilization of 2 ovum by 2 sperms
B) fertilization of abnormal 2n ova with normal sperm
C) Fertilization of normal ova with normal sperm
D) Non disjunction

2. A 10-year-old boy came to the genetic clinic complaining of a genetic disease. Family history revealed a similarly affected sister. His father and paternal grandmother have moderate manifestations. What is the mode of inheritance?

A)Autosomal Dominant
B) X-linked dominant
C) Mitochondrial
D) Holandric

3.Which of the following is an example of matrilineal inheritance?
A) MELAS (mitochondrial encephalopathy, lactic acidosis, strok like episodes)
B) Cystic Fibrosis
C) Achondroplasia
D) Rett Syndrome

4. What is the indication when a normal parent has a child with malformations and an unbalanced chromosomal abnormality?
A) Look for deletion from one of the parents
B) Look for duplication from one of the parents
C) Look for balanced translocation
D) Look for unbalanced translocation

1	2	3	4
B	A	A	D

MCQ

Year : MED21

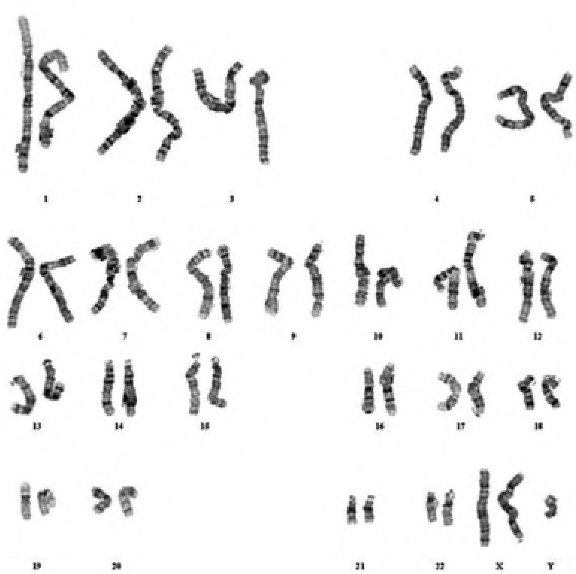
1.Which of the following describes the presence of two types of mitochondrial DNA (normal and abnormal)?

A) Homoplasmy
B) Heterozygous
C) Heteroplasmy
D) Homozygous

2. What is the terminology used to describe thread like structures of DNA and proteins in the nucelus of cells that contain genetic material or DNA?

A) Gene
B) Chromosome
C) Chromatid
D) Centromere

3.What is the diagnosis of the following karyogram?



A) Turner Syndrome
B) Klinefelter syndrome
C) Down syndrome
D) Cri du chat syndrome

4.What is the chromosome that is made of the long arms of two acrocentric chromosomes?

A) Derivative chromosome
B) Metacentric chromosome
C) Telocentric chromosome
D) Homologous chromosome

1	2	3	4
C	B	B	A

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ

Year : MED21

1.Group of signs and symptoms occur at the same time?
A) Sickness
B) Disease
C) Infection
D) Syndrome

2. Between adenine and guanine transformation?
A) Transition (mutation)
B) Transverse
C) Transcription
D) Translation

3.Risk of having an affected child with this syndrome increase in?
A) young maternal age
B) Maternal age below 30
C) Maternal age above 25
D) Maternal age above 45

4.A male (46 chromosomes) is affected with an X-linked dominant disorder and wants to get married. What is the recurrence risk for his children?
A) All daughters affected, none of the sons affected
B) All sons affected, none of the daughters affected
C) 50% risk for each pregnancy
D) Only firstborn daughter affected

1	2	3	4
D	A	D	A

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ

Year :
MED21

1.A 10-year-old boy with a genetic disorder comes to the clinic. He has a similarly affected sister. His father and paternal grandmother are also affected. What is the recurrence risk?

A) 25% for each pregnancy
B) 50% for each pregnancy
C) All children will be affected
D) Only male children will be affected

1
B

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ

Year:
MED18

1- Mode of inheritance of neurofibromatosis is characterized by all the following EXCEPT:

A-male to male transmission is a key feature
B-males and females are equally affected
C-females are obligate carriers
D-variable expression of symptoms and signs in considered

2- Which of the following is true about mitochondrial DNA

A. codes for 70 genes
B. It is circular, single stranded DNA
C. inherited almost exclusively from the oocyte
D. type of mendelian inheritance

3- Marfan syndrome is an example of?

A. Homoplasmy
B. Hetroplasmy
C. pleiotropy
D. polyploidy

4- A male with deletion on chromosome 15 of long arm region 1 band 3.

A. 46, XY, 15 q13
B.46, XY, del 15 q13
C.46, XY, dup 15 q13
D. 46, XY der 15 q13

1	2	3	4
C	C	C	B

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ

Year: MED18

5- All the following are trisomy's except:

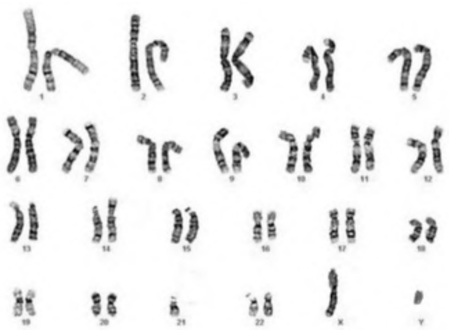
A. Patau's syndrome
B.Downs syndrome
C. Turner syndrome
D. Edward syndrome

6**- An alteration in DNA polymerase lacks exonuclease 5'-3' which results in**:

A. an increase in mutation
B. a decrease in replication
C.decrease in RNA primer removal

7**- What is risk of having this individual down syndrome child?**

A- 100%
B- 25%
C- 1-3 %
D- 10-15%



8- Ahlam is a 47-year-old teacher who has breast cancer, it was found that she has a mutation in the BRCA2 gene. Which of the following is most likely defected?

A.nucleotide excision repair
B. post replication repair
C. mismatch repair

5	6	7	8
C	C	C	A

في حال وجود أي خطأ أو تعديل الرجاء
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والسؤال والإجابة الصحيحة إلى الجروب
الآتي : {اضغط هنا}

MCQ

Year: MED18

9- **Sickle cell anemia is an example of what type of mutation**

A. frame shift
B. Missense
C. Insertion

10**- One of the following enzymes involved in base excision repair?**

A- DNA glycosylase
B-AP endonuclease
C-AP exonuclease
D-reverse transcriptase

1**1- It has been recently determined that the gene for duchene muscular dystrophy (DMD) is over 2000 kb in length,however the mRNA produced by this gene is only about 14 kb.what is a likely cause of this discrepancy?**

A-spliced out of the exons during mRNA processing
B-spliced out of the introns during mRNA processing
C-there are more amino acids coded for the DNA rather than for mRNA
D-when the mRNA is produced, it is highly folded and therefore less long

12-4. .... involves 2 Acrocentric chromosomes fuse near the centromere and loss their short arm
A. isochromosome.
B. chromosomal inversion.
C. reciprocal translocation.
D. Robertsonian translocation

9	10	11	12
B	A	B	D

في حال وجود أي خطأ أو تعديل الرجاء إرسال
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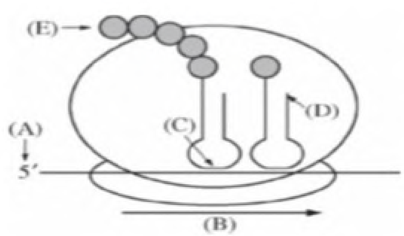
والإجابة الصحيحة إلى الجروب الآتي :
{اضغط هنا}

MCQ

Year:
MED18

13- **Which of the following is correct about Autosomal recessive trait?**
A. Only manifest in male
B. Manifest in heterozygous
C. Recurrence risk is 50% in females carriers
D. Skip generations



14- In eukaryotes translation during the elongation stage in this diagram which of the following is NOT CORRECT?
A. the cap end of mRNA
B. the direction of the ribosome
C. anticodon of tRNA
D. 3'end of the tRNA
E. amino terminus of polypeptide chain

15- 10. Defect in Mismatch repair will lead:
A. xeroderma
B. non-polyposis colorectal cancer
C. Nijmegen syndrome
D. breast cancer

16- **What could be the effect of a coding sequence of 114 nucleotides including start and stop codons that has an insertion of 9 nucleotides at position** 82?

A. Insertion of 3 amino acids
B. No effect
C. First 68 nucleotides are affected
D. Last 11 nucleotides are affected

13	14	15	16
D	D	B	A

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إلى الجروب الآتي : {اضغط هنا}

MCQ

Year:
MED18

17- Which one is true for Huntington disease:

A. CCG, Normal range 4 - 39 repeats.
B. CTG, Normal range 5 - 37 repeats.
C-CAG, Disease range more than 40 repeats.
D- CGG, Disease range 200 - 900 repeats.

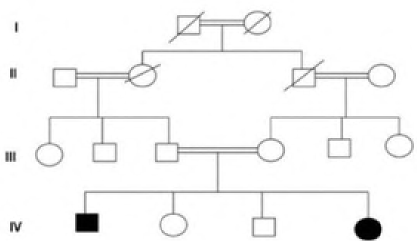
18- Genomic imprinting considered if:

A.The individual inherits both homologus chromosome from one parent
B. The normal individual has only one transcriptional active gene
C. Individual have more then on cell type
D. onset at earlier age in the offspring than in parent

19- The following karyotype is for:

A. Normal Male
B. Normal Female
C. Turner Syndrome
D. Klinefelter Syndrome

20- Regarding to the pedigree, which of the following statement is correct?



Answer: family history of sickle cell anemia

17	18	19	20
C	B	B	A

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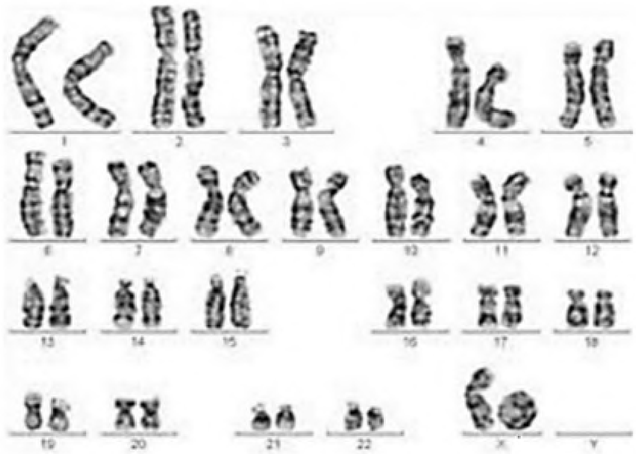
والسؤال والإجابة الصحيحة إلى الجروب
الآتي : {اضغط هنا}

21- what is the following karyotype:



A. 46, XX,del (4p)
B. 46, X,isoXq
C. 46, X,(X)
D. 46, XX

21			
C			

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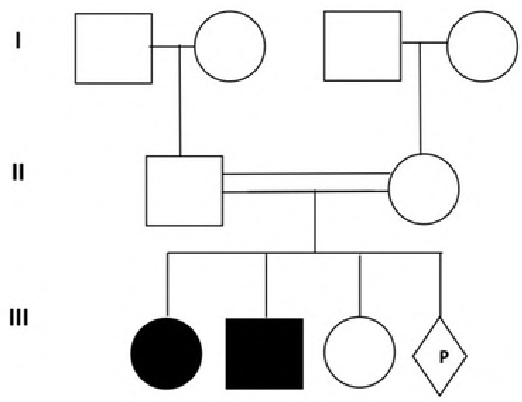
والسؤال والإجابة الصحيحة إلى الجروب
الآتي : {اضغط هنا}

Short Essay

Year: MED18

Consanguineous marriage parent have Female daughter suffering from sickle cell anemia and have a brother affected and a sister normal and the mother is pregnant.

A- Draw three generation pedigree.



B- The mode of inheritance?
Answer: Autosomal recessive

C- The risk recurrence of the expected child?
Answer: 25% per each pregnancy

2- If a couple (Ahmed and Fatimah) came to the genetics clinic, and they are expecting a new baby. After analyzing their chromosomes, it showed that Ahmed has the following nomenclature 46, XY, der (21;21), +21.

A- **What type of chromosome abnormality does he have?**
Answer: Robertsonian translocation

B- **Do you agree that Ahmed has balanced chromosomal abnormality?**
Answer: No

C- What type of chromosomal anomalies their baby will have and the percentage?
Answer: Down syndrome, 100% per each pregnancy

MCQ

Year : MED15

1. Consanguineous parents have a high risk of developing what kind of disease ?
A) cystic fibrosis
B) Marfan syndrome
C) hemophilia
D)Neurofibromatosis

2. A karyotype of female Down syndrome:
A) 46,XX,rob(14;21)+21
B) 47,XXY
C) 46,XX,rob(13;15)+21
D) 46,XY,der(12;21)+12
E) 47,XX,+18

3. A 6-year-old female presents with hyperphagia and short stature. Clinical examination reveals obesity and dysmorphic features, including almond-shaped eyes and small hands and feet. What is the most likely diagnosis?
A) Angelman syndrome
B) Prader–Willi syndrome
C) Turner syndrome
D) Klinefelter syndrome

4.The addition of a poly-A tail to mRNA serves which of the following functions?
A) Protects mRNA from degradation
B) Facilitates the passage of mRNA through the nucleus
C) Binds to the promoter
D) Enhances ribosomal binding and translation efficiency

1	2	3	4
A	A	B	A

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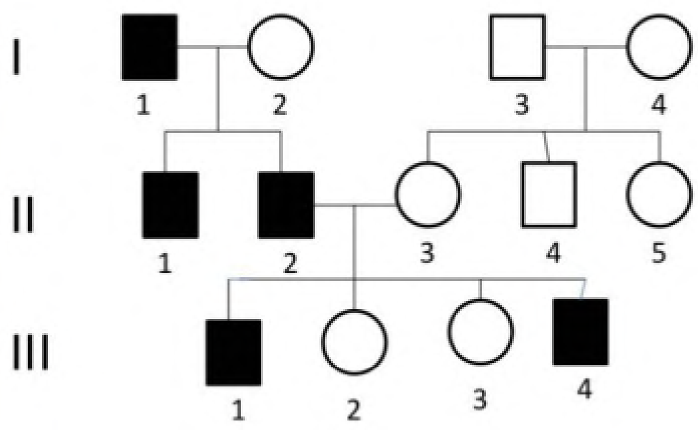
MCQ

Year : MED15

1. One of the following disorders is caused by a genetic imprinting defect leading to developmental delay, seizures, and a happy demeanor:
A) Sickle cell anemia
B) Angelman syndrome
C) Huntington disease
D) Neurofibromatosis

2.What's the mode of inheritance of this pedigree?
A- Hollandaric
B- autosomal recessive
C- X—linked recessive
D- autosomal dominent
E- x-linked dominant



3. An affected male transmits the disease to all of his daughters but none of his sons. This pattern is most consistent with:
A) X-linked recessive disorders
B) X-linked dominant disorders
C) Mitochondrial disorders
D) Autosomal dominant disorders

4.A male presents with dysmorphic facial features, almond-shaped eyes, small hands and feet, hyperphagia, short stature, and obesity. What is the most likely diagnosis?
A) Prader–Willi syndrome
B) Down syndrome
C) Turner syndrome
D) Klinefelter syndrome

1	2	3	4
B	A	B	A

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ

Year : MED15

1. Polydactyly is a unique feature of which of the following disorders?
A) Patau syndrome
B) Cri du chat syndrome
C) Achondroplasia
D) Down syndrome

2.A karyotype shows a ring X chromosome. Identify the disorder:
A) Down syndrome
B) Klinefelter syndrome
C) Turner syndrome
D) Patau syndrome

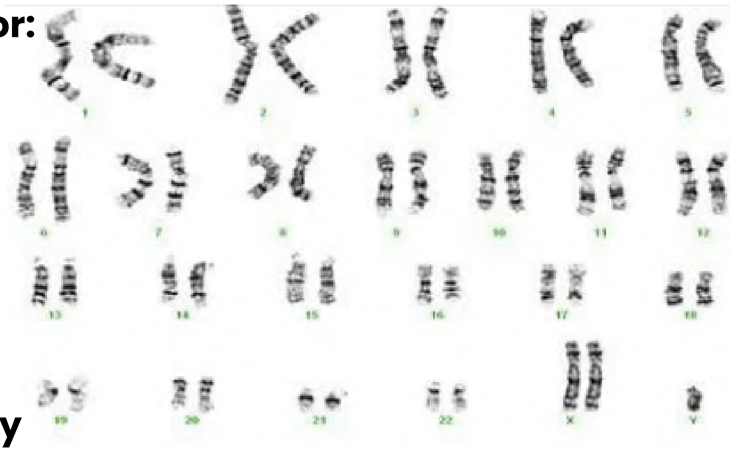
3. **Which of the following statements is not correct regarding the genetic code?**
A) It is a triple code
B) It reads as a series of nucleotides
C) The codon can undergo mutation
D) It is not universal for all organisms
E) There is more than one codon for an amino acid

1	2	3
A	C	D

MCQ

Year :
MED15

1. The following karyotype is for:



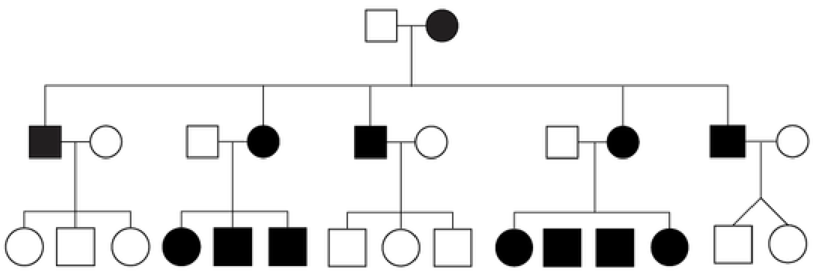
A- Klinefelter Syndrome
B- Patau Syndrome
C- Cri du chat Syndrome
D- Turner Syndrome
E- Down Syndrome

2.Which of the following syndromes is characterized by pleiotropy?

A) Marfan syndrome
B) Turner syndrome
C) Down syndrome
D) Prader–Willi syndrome

3.Which of the following is most likely to be a feature of the disorder shown in the pedigree?

A) Maternal inheritance
B) Paternal inheritance
C) Autosomal dominant inheritance
D) X-linked recessive inheritance



1	2	3
A	A	A

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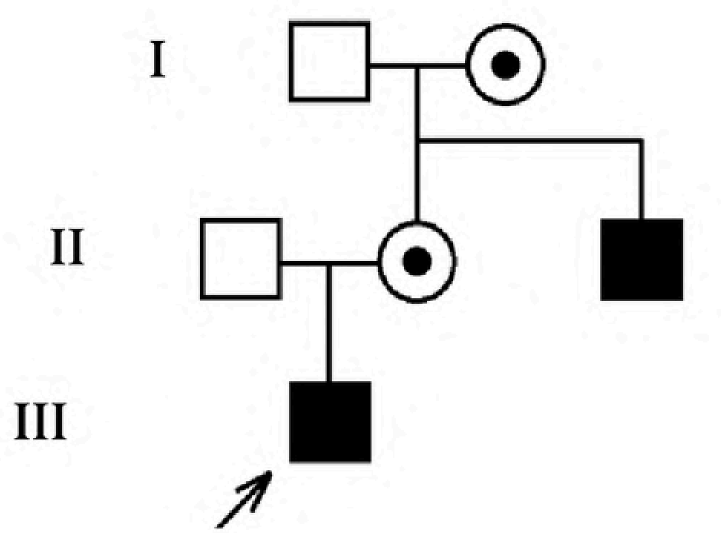
Short essay

Year : MED15

8 Years old male patient presented with rapid falling and hardly stand up by his own , clinical examination reveals muscle weakness repeated falling . pseudo hypertrophy of calf muscle there is also a maternal uncle that share the same symptoms

A.Draw the 3 generations pedigree

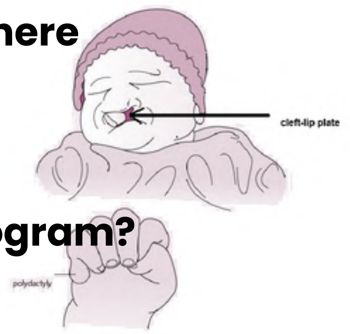


B. what is your diagnosis ?
dunchene muscular dystrophy
C.mode of inheritance ?
x-linked recessive
D. recurrence risk ?
Answer: 50% (1/2) affected male 50% (1/2) carrier female

Short essay

Year : MED15

A. Name the 2 things that are arrowed in here



Answer: 1-cleft-lip palate 2-polydactyly

B. What is your diagnosis?

Answer: Patau's syndrome

C.what is the expected result of the karyogram?

Answer: trisomy 13

A child presents with anemia and episodic pain in the lower back, legs, abdomen, and chest, with periods of remission. Episodes tend to recur in the same areas. Her parents are first cousins, and there is a family history of similar symptoms.

A. Diagnosis:

Answer: Sickle cell anemia

B. Mode of inheritance

Answer: Autosomal recessive

C. Recurrence risk:

25% per pregnancy

MCQ + Short

Year : MED15

1.There was a family which came in the clinic with a child diagnosed with sickle cell anemia what is the recurrence risk for this type of disease ?
A. 25% carrier
B. 50% Carrier
C. 100% Affected
D. 50% Affected

2.A father is affected with an X-linked dominant disorder. Which of the following is true regarding this disease?
A) Females are affected as well as males
B) The father transmits the disorder to all of his sons
C) Only males are affected
D) The disorder can skip generations

3.Diploid definition -cellular condition where each chromosome type is represented by two homologous chromosomes.

4.Mosaicism Definition . -Two or more cll lines that differ in their genetic constitution but are derived from a single zygote.

1	2
B	A

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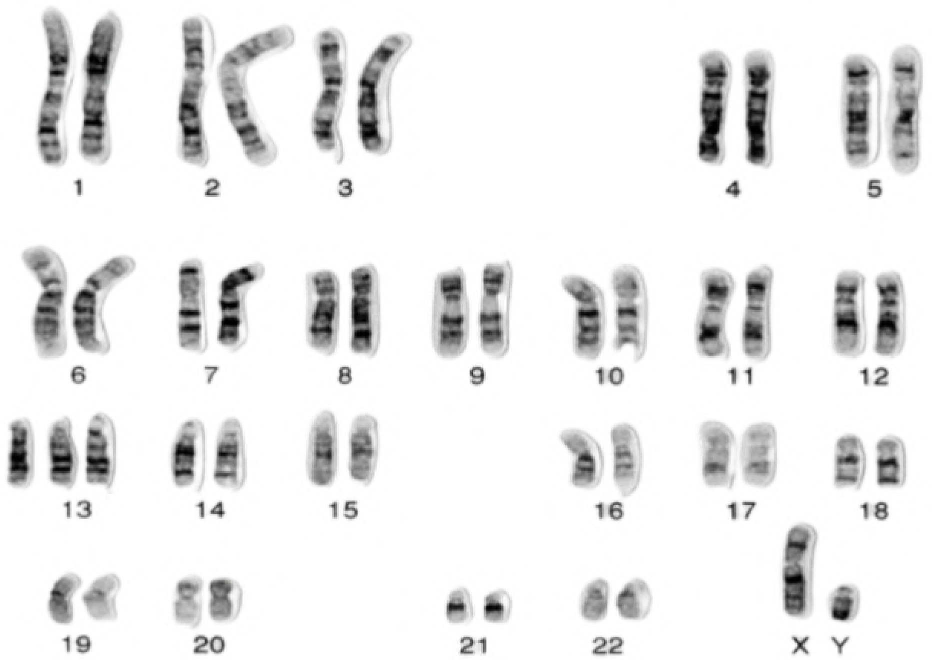
Short answer

Year : MED15

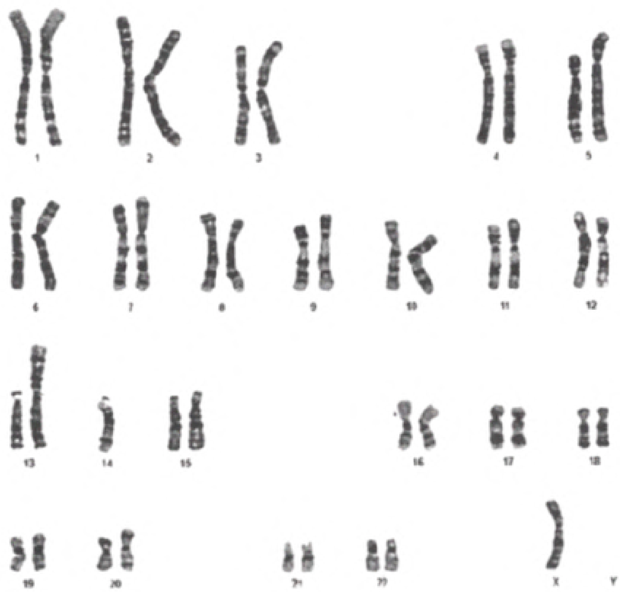
According to the following karyotype give your response. (3 Marks)

A.



Answer: Trisomy 13 (Patau's syndrome) 47,XY,+13

B.



Answer: Turner's syndrome 45,X

Short answer

Year : MED15

According to the following karyotype give your response. (3 Marks)

C.

Answer: Cri du Chat Syndrome 46,XY,del (5)(p15)
5p- syndrome

True or false

Determine whether the following statements are True or False:

A) Chromosomes can be identified by their thickness.

False (size, banding pattern and centromere position)

B) A karyotype showing 46,XX,der(14;21)+21 is considered a Robertsonian translocation.

True

C) 47,XXY is a numerical chromosomal abnormality.

True

D) Mitosis is the process of nuclear division that occurs during the final stage of gamete formation.

False (Meiosis)

MCQ

Year :
MED15

1.A female was in the clinic with her daughter, her daughter was 47 t (14;21)+21 Unfortunately her mother was also diagnosed by 45 t (14 ; 21) What is the percentage for the next birth to be affected with the same effect:

A. 0.3 %
B. 3 -5 %
C. 10 -15 %
D. 100%

2.A Karyotype shoed the following 46,XY, t(4;6), (p12;q14) How is it named?

A. Robersonian translocation between 4 and 6.
B. Reciprocal translocation between chromosome number 4 and 6.
C. Reciprocal translocation between chromosome 12 and 14

3.Fragile X Syndrome is caused by:

A. Expansion of CAG repeats > 39
B. Expansion on CTG repeats > 39
C. Expansion of CGG repeats > 200
D. Expansion of CTG repeats > 200

4.This karyotype is diagnosed for:

A. cri du chat
B. wolf syndrome
C. down syndrome
D. Edward syndrome
E. patau syndrome

1	2	3	4
c	B	C	A

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الصحيحة إلى الجروب الآتي : {اضغط هنا}

MCQ + Short

Year : MED15

1.Give the definition of homologous chromosome:

A) A chromosome of a different size and shape which carries different types of genes
B) A chromosome of the same size and shape which carries the same type of genes
C) A chromosome of the same size and shape which carries different types of genes
D) A chromosome of a different size and shape which carries the same type of genes

2.Give the definition of Ring chromosome:

A) A numerical chromosomal abnormality in which two chromosomes have been translocated to another one to form a ring
B) A structurally abnormal chromosome in which two chromosomes have been united with each other to form a ring
C) A numerical chromosomal abnormality in which a new chromosome has been united with another one to form a ring
D) A structurally abnormal chromosome in which the ends of each chromosome arm have been lost and the broken arms have been reunited to form a ring

3.Mustafa and Fatima are a couple seeking genetic counseling. They have a male baby with a chromosomal aberration: 46,XY,t(21;21)+21. Cytogenetic analysis for Fatima shows the same Robertsonian translocation. What is the recurrence risk in the next pregnancy for having a similarly affected baby?

A) 1%
B) 5%
C) 5–10%
D) 10–15%
E) 100%

When a chromosome divides transversely at the centromere instead of along its length, it results in a chromosome with two identical short arms or two identical long arms. What is this type of chromosome called?

Answer:isochromosome

Concerning the centromere position, chromosome 3 is considered a _____ chromosome.

Answer:metacentric

1	2	3
B	D	E

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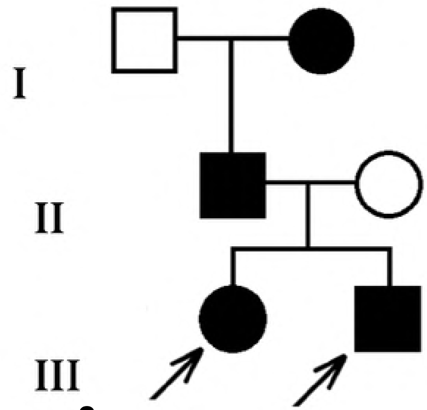
Short essay

Year : MED15

A Family with a 6 years old girl and a 4 years old boy go to the doctor, both with a well known genetic disease. The Father and Paternal grandmother manifest the same disease

1. Draw a pedigree for this family



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2.What is the mode of inheritance for this disease?

Answer: Autosomal dominant

3.What is the recurrence risk ?

Answer: 50% per pregnancy

4.Mention 2 criteria for this mode of inheritance.

Answer: - Does not skip generations -Both females and males are equally affected

Nada and Mahmoud are consanguineous parents and came to the genetic clinic with a child that has sickle cell anemia

1.What is the recurrence risk per pregnancy?

25% per pregnancy

2.What is the risk percentage for a carrier ?

50% per pregnancy

Second mid

1.During dysmorphology assessment, a child is found to have a shortened anteroposterior dimension of the skull compared to its width. What is the correct term for this finding?

A. Dolichocephaly
B. Plagiocephaly
C. Brachycephaly
D. Trigonocephaly

2.Which term describes the presence of a small mandible?

A. Macrognathia
B. Retrognathia
C. Micrognathia
D. Prognathia

3.A newborn is diagnosed with an isolated congenital heart defect. Which type of birth defect does this represent?

A. Dysplasia
B. Disruption
C. Malformation
D. Deformation

4.Which of the following birth defects is considered a disruption?

A. Congenital heart defect
B. Neural tube defect
C. Cleft lip and palate
D. Amniotic band syndrome

1	2	3	4
C	C	C	D

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MED25

1.Certain malformations tend to occur together more frequently than expected by chance. Which of the following best represents this concept?

A. Down syndrome
B. VACTERL association
C. Ectodermal dysplasia
D. Cleft lip and palate

2.A newborn presents with vertebral anomalies, anal atresia, tracheoesophageal fistula, and renal abnormalities. What is the most likely diagnosis?

A. Potter sequence
B. Van der Woude syndrome
C. VATER association
D. Ectodermal dysplasia

3.Which DNA repair mechanism involves the enzyme DNA glycosylase?

A. Mismatch repair
B. Nucleotide excision repair
C. Base excision repair
D. Post-replication repair

4.Which disorder is commonly associated with mismatch repair defects due to incorrect base pairing during DNA replication?

A. Bloom syndrome
B. Xeroderma pigmentosum
C. Colorectal cancer
D. Thalassemia

1	2	3	4
B	C	C	C

MED25

1.Which of the following best describes a nonsense mutation?

A. A substitution causing a different amino acid
B. A substitution creating a premature stop codon
C. Deletion of three nucleotides
D. Insertion without frameshift

2.Which of the following variants is an example of a missense mutation?

A. p.Gly542*
B. c.489+1G>T
C. p.Arg117His
D. p.Phe508del

3.Which of the following variants is an example of a nonsense mutation?

A. p.Arg117His
B. p.Gly542*
C. p.Phe508del
D. c.489+1G>T

4.Which of the following disorders is most commonly caused by loss of function of the protein product?

A. Huntington disease
B. Achondroplasia
C. Marfan syndrome
D. Familial hypercholesterolemia

1	2	3	4
B	C	B	D

MED25

1.When a mutant gene product interferes with the function of the normal protein in a heterozygous state, this is known as:

A. Haploinsufficiency
B. Genomic imprinting
C. Dominant-negative effect
D. Uniparental disomy

2.Which codons function as stop codons and terminate translation?

A. AUG and UAA
B. UUU, UAA, and UAG
C. AUG, UAA, and UAG
D. UGA, UAA, and UAG

3.Which sequence correctly describes the major DNA repair mechanism?

A. Cleavage → Removal → Insertion → Sealing
B. Removal → Cleavage → Insertion → Sealing
C. Cleavage → Insertion → Removal → Sealing
D. Cleavage → Removal → Sealing → Insertion

4.Which factor most strongly influences recurrence risk in multifactorial disorders?

A. Father's age at conception
B. Number of unaffected siblings
C. Degree of relationship to the index case
D. Presence of de novo variants

1	2	3	4
C	D	A	C

MED25

1.A child with Marfan syndrome is started on medication to reduce the risk of aortic root dilation. Which class of drugs is most commonly used?

A. Corticosteroids
B. Beta blockers
C. Testosterone replacement
D. Folic acid supplementation

2.Chromosome analysis would be most helpful in diagnosing which of the following disorders?

A. Neurofibromatosis type 1
B. Cystic fibrosis
C. Klinefelter syndrome
D. Duchenne muscular dystrophy

3.Which term describes the presence of an extra digit?

A. Syndactyly
B. Oligodactyly
C. Arachnodactyly
D. Polydactyly

4.A child has an increased distance between the eyes. Which term best describes this finding?

A. Upslanting palpebral fissures
B. Hypotelorism
C. Hypertelorism
D. Anophthalmia

1	2	3	4
B	C	D	C

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MED25

1.A 2-year-old child has developmental delay and multiple congenital anomalies. Chromosome analysis and microarray are negative, and no specific syndrome is suspected. What is the most appropriate next test?

A. FISH
B. MLPA
C. Sanger sequencing
D. Whole exome sequencing

2.A woman with breast cancer is found to carry a pathogenic CHEK2 variant. Which test is most appropriate to screen her siblings for the same mutation?

A. Microarray
B. MLPA
C. Whole exome sequencing
D. Sanger sequencing

3.A 14-year-old boy with hypotonia and upslanting palpebral fissures is diagnosed with Down syndrome. What is the most likely underlying mechanism?

A. Gene deletion
B. Gene amplification
C. Monosomy
D. Maternal nondisjunction causing trisomy

4.A male newborn with vertebral anomalies, anal atresia, tracheoesophageal fistula, and radial ray defects has normal WES and microarray results. What is the most likely diagnosis?

A. Potter sequence
B. VATER association
C. Van der Woude syndrome
D. Ectodermal dysplasia

1	2	3	4
D	D	D	B

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MED25

1.A 4-year-old girl has short stature, webbed neck, pulmonary valve stenosis, and downslanting palpebral fissures. Which genetic abnormality is most commonly associated with this condition?

A. Gain-of-function mutation in PTPN11
B. Loss-of-function mutation in PTPN11
C. Gain-of-function mutation in NF1
D. Loss-of-function mutation in NF1

2.A 7-year-old boy presents with calf pseudohypertrophy and positive Gowers sign. Molecular testing reveals a deletion causing loss of dystrophin. Which therapy may help restore the reading frame?

A. SMN1 gene replacement
B. DMD exon skipping therapy
C. SMN2 exon inclusion therapy
D. FGFR3 inhibition with CNP analog

3.A neonate presents with macrosomia, macroglossia, umbilical hernia, and neonatal hypoglycemia. Which molecular mechanism is most likely responsible?

A. FGFR3 loss-of-function mutation
B. Maternal UPD of chromosome 7
C. 22q11.2 deletion
D. Abnormal imprinting at chromosome 11p15.5

4.A child with neurofibromatosis type 1 has café-au-lait spots and optic glioma. His mother has Lisch nodules and mild hearing loss. What is the recurrence risk in the next pregnancy?

A. 25%
B. 50%
C. 75%
D. 1–3%

1	2	3	4
A	B	D	B

MED25

1.A child is diagnosed with neurofibromatosis type 1 (NF1) and presents with multiple café-au-lait spots and optic glioma. His mother also has NF1 but only mild hearing impairment and Lisch nodules. Which genetic concept best explains the difference in clinical severity between family members?

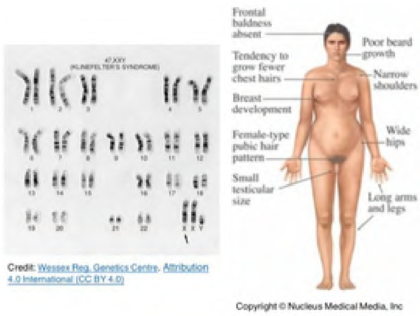
A. Anticipation
B. Variable expressivity
C. Incomplete penetrance
D. Genomic imprinting

1
B

1-Identify the syndrome shown in the image?

Ans: Marfan syndrome.

2-Identify the syndrome shown in the image?



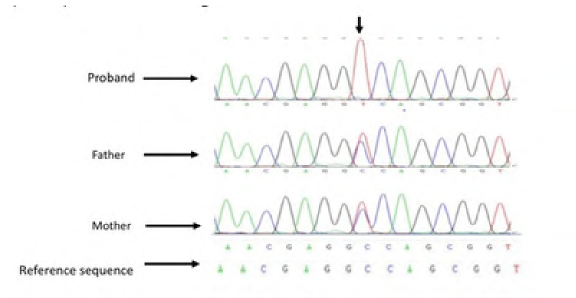
Ans: Klinefelter syndrome.

3-What is the mode of inheritance of the condition shown in the image?

Ans: X-linked recessive

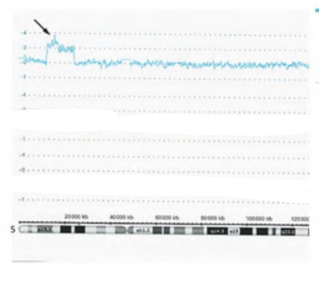
MED25

4-interpret the following result



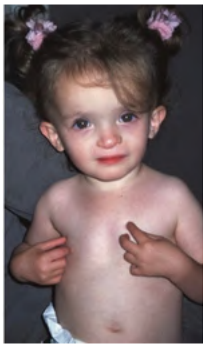
Ans: Homozygous mutant

5- Interpret the following result and identify the affected chromosomal arm



Ans: Heterozygous duplication of the p arm

6-Identify the syndrome shown in the image.



Ans: Silver-Russell syndrome

MED25

7-A newborn presents with macroglossia, abdominal wall defect, and overgrowth features. Which genetic test is most appropriate for diagnosis?

Ans: Methylation analysis

8-Identify the syndrome shown in the image.

Ans: Achondroplasia

9-Identify the syndrome shown in the image.

Ans: Neurofibromatosis type 1

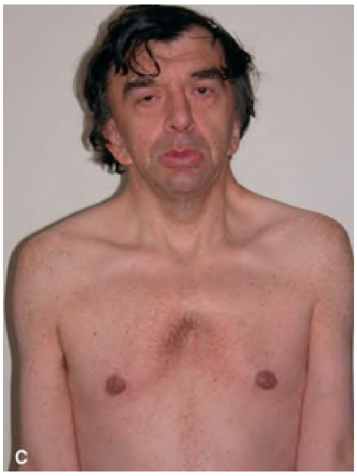
JACKAROO 2

SHORT ESSAY Q's:

1) Explain mRNA splicing mutations

2) Marfan Syndrome and gene affected

3) What is the diagnosis and gene affected



4) What is the diagnosis and give two clinical features

JACKAROO 2

1) Elevated amniotic alpha fetoprotein (AFP) has been detected prenatally in a patient. To reduce the risk of developing the consequence shown in the photo, what could be given?

a. Vitamin B12
b. Folic acid
c. IgA
d. Immunosuppressant

2) An 18-month old is presented with muscle weakness and frequent falls. The doctor suspects a point mutation on the SMN1 gene. How can we confirm this?

a. MLPA
b. Chromosomal Microarray
c. Sanger Sequencing

3) Responsible for correction of thymidine dimer:

a. BER
b. NER
c. MMR

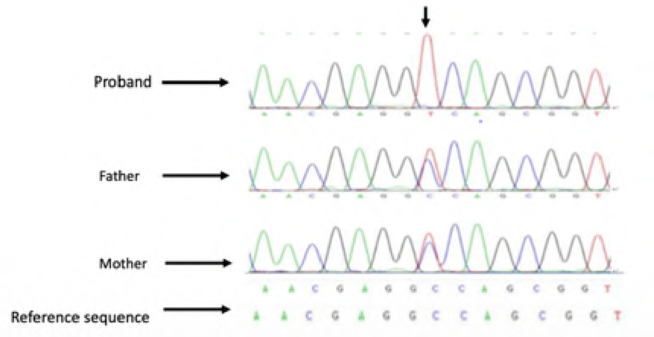
1	2	3	4
b	c	b	

JACKAROO 2

1) What method can be used to detect DMD?

a. MLPA
b. FISH
c. Microarray

2) The following Sanger Sequencing has been done for healthy parents and an affected son. What is the mode of transmission?



a. X-linked
b. autosomal dominant
c. autosomal recessive
d. spontaneous mutation

3) A patient is presented with long stature and arms, aortic root dilation, scoliosis, and pes planus (flat feet). What is the affected gene?

a. FMN 1
b. FBN 1
c. FVN 1

4) How to detect anemia carrier:

a. CBC
b. Electrophoresis
c. Bone marrow biopsy

1	2	3	4
a	c	b	b

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JACKAROO 2

1) Which of the following is associated with Noonan Syndrome?

a. bleeding diseases
b. scoliosis
c. short stature

2) Cri-du-chat Syndrome is caused by:

a. duplication in 5p
b. deletion in 5p
c. duplication in 5q
d. deletion in 5q

1	2	3	4
a	b		

Genetics Final Season

15) Failure to thrive and malnutrition as result of chronic diarrhea and malabsorption due to pancreatic insufficiency, are clinical features of:

A. CFTR

B. NF1

16) primary and intrinsic…

A. malformation

20) Which of the following DNA repair mechanisms can cause mistakes?

A. homologous recombination

B. non homologous end joining

C. direct reversal

D. mismatch repair

21) Which type of mutation is controversial to genetic therapy?

A. somatic mutation

B. germline Mutation

C. ex vivo

D. anti-seneal mutation

15	16	20	21
A	A	B	B

Genetics Final Season

22) What is the term used to describe the vertical groove between the base of the nose and the border of the upper lip:

A. Philtrum
B. Hard palate
C. Soft palate
D. Pinna

24) The following chart shows the stature of a 13 years old girl with primary amenorrhea, congenital heart disease and webbed neck. What is the abnormality?

A. Aberration in X chromosome
B. Autosomal recessive
C. Autosomal dominant

25) which of the following terms describes the abnormal mechanical force which distorts an otherwise normal structure.

A. malformation
B. Disruption
C. Deformation
D. Dysplasia

26) What is the best description of disruption:

A. primary and extrinsic
B. Secondary and extrinsic
C. Primary and intrinsic

22	24	25	26
A	A	C	B

Med 22

Genetics Final Season

30) Sara is a 30 years old female who came to a genetic counseling with a history of breast cancer in family. Her mother had breast cancer caused by BRCA1. Which of the following tests is appropriate for sara?

A. chromosome microarray
B. MLPA
C. sanger sequencing
D. whole exome sequences

33) A 47-year-old female patient is referred to the genetic clinic, tumor in her right colon. Her father died at the age of 55 from colon cancer, her paternal aunt had endometrial cancer. What is the best technique to confirm her condition?

A. Gene panel
B. Chromosomal analysis
C. Search for MLH mutation
D. Sanger sequencing

40) A couple came to genetic clinic, they have a 12 years old girl presented with short stature, enlarged head and rhizomelic shortening of long bone. How do you confirm the diagnosis?

A. DNA analysis for FGFR3
B. chromosomal analysis
C. FBN1 dna analysis
D. glycoproteins

30	33	40	
C	A	A	

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42) Which of the following is used in base excision repair mechanism?
A. ATM
B. O6-methylguanine-DNA-methyltransferase
C. MLH
D. DNA glycosylase

44) A family with two affected boys presented to the genetic clinic. The two brothers have muscle weakness and a positive gower's sign. Family history revealed a maternal uncle that is wheelchair dependent and has dilated myocardial myopathy. What additional manifestation can be observed in this condition?
A. spinal compression
B. Calf pseudohypertrophy
C. Macrocephaly
D. Microcephaly

45)A family with two affected boys presented to the genetic clinic. The two brothers have muscle weakness and a positive gower's sign. Family history revealed a maternal uncle that is wheelchair dependent and has dilated myocardial myopathy. Which of the following lab results will be expected?
A. High creatine kinase level.
B. Low creatine kinase level.
C. Normal creatine kinase level.
D. No change to creatine kinase level.

42	44	45	
D	B	A	

Genetics Final Season

46) A family with two affected boys presented to the genetic clinic. The two brothers have muscle weakness and a positive gower's sign. Family history revealed a maternal uncle that is wheelchair dependent and has dilated myocardial myopathy. Which of the following type of gene is most likely to be responsible for this history?

A. Dystrophin encoding gene
B. Neurofibromin encoding gene
C. Fibrillin-1 encoding gene
D. Calcium channels encoding gene

47) A family with two affected boys presented to the genetic clinic. The two brothers have muscle weakness and a positive gower's sign. Family history revealed a maternal uncle that is wheelchair dependent and has dilated myocardial myopathy. Which of the following genetic abnormalities is most likely present in this family?

A. Gene deletion
B. Gene duplication
C. chromosome deletion
D. chromosome Duplication

46	47		
A	A		

Genetics Final Season

56) What best describes short fingers?
A. clindoductly
B. brachoductly

57) What is a major anomaly of Marfan syndrome?
A. Arachondactylyl
B. Aortic aneurysm
C. Pes planus
D. Short stature

59) Which of the following is considered as a major defect?
A. Microcephaly
B. Lacrimal duct stenosis
C. Epicanthic fold
D. Brushfield spots of the iris

60) A female presented in the NICU with heart and renal abnormalities. After examination: microcephaly microphthalmia, hypertelorism, postaxial polydactyly.
What is the provisional diagnosis?
A. patau syndrome
B. down syndrome

56	57	59	60
B	B	A	A

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Genetics Final Season

63) UV light cause:

A. pyrimidine dimers
B. purine dimers
C. Single dna strand break
D. Double dna strand break

64) Which of the following involves the mechanism of triple repeat sequence in Huntington's disease?

A. Gain of function
B. Increased inhibition of expression

69) best description of missense mutation

A. Result in coding for a different amino acid and the synthesis of an altered protein

70) what's correct regarding neural tube defect

A. Anencephaly results from defect in the lower end
B. It risk can be reduced by the intake of Periconceptional folic acid
C. RR from a parent is 10%

63	64	69	70
A	A	A	B

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Genetics Final Season

72) what is the best genetic testing for chromosomal duplication <3 mega base pair

A. Microarray

B. Chromosomal analysis

C. MLPA

73) What is the term used to describe microstomia?

A. small tongue

B. Small mouth

C. Small tongue

72	73		
A	B		

Seasons 2

1) Which of the following is considered as major anomaly?

A. Lip pits
B. Anencephaly
C. Epicanthic fold

2) Which of the following vitamins could be used to treat homocysteinuria?

A. Vit B1
B. Vit B6
C. Vit C
D. Vit A

3) Which of the following is caused by abnormal organization of tissue?

A. Ectodermal dysplasia
B. Spina bifida
C. Cleft palate
D. Talipes

4) Hurler and Hunter can present similarly, which feature can distinguish between them?

A. Hepatosplenomegaly
B. Corneal clouding
C. Coarse facial features
D. Developmental delay

5) Which of the following disorders develops arachnodactyly of fingers and toes?

A. Marfan
B. Cystic fibrosis

1	2	3	4	5
B	B	A	B	A

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Seasons 2

6) The family came with their 7-year-old girl and 5-year-old boy, complaining of chronic pulmonary disease, failure to thrive with malnutrition and chronic diarrhea due to pancreatic insufficiency, what is the mode of inheritance?

A) AR
B) AD
C) XL
D) single gene

7) What would the clinical test results indicate to the family who came with their 7-year-old girl and 5-year-old y.o boy complaining of chronic pulmonary disease, failure to thrive with malnutrition, and chronic diarrhea due to pancreatic insufficiency?

A) High sweat chloride
B) Negative sweat chloride
C) Low sweat chloride
D) Normal sweat chloride

8) What is the gene responsible for the family that came with their 7-year-old girl and 5-year-old boy complaining of chronic pulmonary disease, failure to thrive with malnutrition, and chronic diarrhea due to pancreatic insufficiency?

A) Chloride channels
B) Calcium channels
C) Sodium channels

6	7	8
A	A	A

Seasons 2

9) Family came with their 7-year-old girl and 5-year-old boy, complaining of chronic pulmonary disease, failure to thrive with malnutrition, and chronic diarrhea due to pancreatic insufficiency, what other clinical feature might be seen?

A) male infertility due to testicular atrophy

B) Male infertility due to congenital bilateral absence of the vas deferens

C) Female infertility due to ovarian failure

10) Which of the following is considered a malformation?

A) Ventricular septal defect

B) Brushfield spots of the iris

C) supernumerary nipple

D) Lip pits

11) What is the main O2 transport protein in humans during the last 7 months of pregnancy in the uterus?

A) HbH

B) HbF

C) Gower I

12) A 3-week-old baby came to the emergency room with vomiting, diarrhea, organomegaly, and loss of consciousness. What is the diagnostic and differential method?

A) Blood glucose

B) Acylcarnitine profile

C) Plasma very long-chain fatty acid

D) Galactose-1-phosphate

9	10	11	12
B	A	B	D

Seasons 2

13) A patient presents with cafe au lait spots and optic nerve glioma. What gene is responsible for this?

A) FBN1

B) NF1

C) FGFR3

14) Which of the following is an X-linked disorder?

A) OTCM

B) PKU

C) Glycogen storage disorder

15) A 3-year-old boy with protruded tongue, saimin crases, and microcephaly. What is the susceptible syndrome?

A) Down syndrome

16) What is the best test for diagnosing the previous syndrome?

A) Cytogenetic test

B) Whole exome sequencing

17) Chromosome analysis can be used for which of the following?

A) Klinefelter syndrome

B) Fragile X syndrome

18) Which of the following is the result of three alpha globin alleles?

A) HbH disease

B) sickle cell disease

13	14	15	16	17	18
B	A	A	A	A	A

Med 18 final

1. 6-year-old boy presented with hypogonadism, obesity, almond shaped eyed. What is the diagnosis?

A- Down syndrome
B- Fragile X
C- Angelman syndrome
D- Prader Willi

2. The enzyme use converted RNA to DNA:

A- Reverse decarboxylase
B- DNA polymerase
C- Reverse transcriptase

3. Male with TB disease use of isoniazid and have severe adverse effect:

A- Increase the dose to control the side effect
B- Continue using without stopping
C- Change the drug
D- take additional drugs

4. A couple came to the genetic clinic with a child having bilateral lip/palate, what is the chance of them having a second child with the same dysmorphogenetic?

A- 5.6%
B- 2%
C- 50%
D- 4%

5. Medication given in first trimester of pregnancy?

A- Streptomycin
B- Folic acid
C- Isoniazid

6. Two parents came to counseling, the have 2 affected children with hemophilia A. They are interested in having a healthy child what you recommend for them?

A- preimplantation genetic diagnosis
B- newborn screening of hemophilia A
C- early diagnosis

1	2	3	4	5	6
D	C	C	D	B	A

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Med 18 final

7. Mutations in BRCA2 gene:

A- Account for 3% to 5% of familial breast cancer
B- account for 100% of familial breast cancer
C- Males with BRCA2 mutation do not have any risk to develop breast cancer
D- Males have an increased risk to develop a prostate cancer

8. What is the Potential Reasons for Pediatric Genetic Counseling?

A- Adult with a family history of a genetic condition.
B- Adult with a strong family history of common adult onset disorders
C- Adult with a strong family history of cancer
D- Child with a family history of a genetic condition.

9. An increase in inhibin-A level in maternal serum, is a diagnosis of?

A- Cystic fibrosis
B- Down syndrome pregnancies
C- Turner syndrome pregnancies
D- Duchenne muscular dystrophy pregnancies

10. A mother that has 1 child with galactosemia is pregnant in her 2nd trimester. What is the test advice?

A- newborn screening of galctosemia
B- parent carrier test of galactosemia
C- ultrasonography

11. According to genetic counseling all the following is true except:

A- Genetic counseling is directive.
B- Any physician can be a counselor.
C- It includes the discussion of the cause of the disease and the risk of transmission.
D- It should be done after genetic testing.

7	8	9	10	11	12
D	D	B	B	B	A

Med 18 final

Which of the following is considered as a major congenital anomaly in Prader-Willi syndrome?
A- Hypogonadism
B- Polydactyly
C- Almond shaped eyes
D- Upward slanting of the eye

13. PCR amplifying DNA by using……… isolated from living bacteria
A- DNA polymerase
B- Restriction enzyme
C- Plasmid
D- Phages

14. What of the following DNA repair mechanisms is a Direct reversal mechanism?
A. Post-replication
B. Excitation
C. Mismatch repair
D. Photoreactivation

15. Mutations in 4 alpha globin alleles is associated with?
A- Hb H
B- Hydrops fetalis
C- Cooley's anemia

16. A 30-year-old male had chronic pulmonary disease and infertility which of the followings could be the diagnosis?
A- Klinefelter syndrome
B- Cystic fibrosis
C- Marfan syndrome
D- Prader Willi Syndrome

17. Which of the following prenatal test doesn't carry a risk of fetal loss?
A- Maternal serum AFP
B- Chronic villus sampling
C- Fetal tissue sampling
D- Amniocentesis

12	13	14	15	11	18
A	A	D	B	C	A

Seasons 2

19) Which of the following disorders is due to a structural change in the globin chain?
A. Sickle cell anemia
B. HbH
C. Delta-beta thalassemia

20)Which of the following has the highest recurrence risk?
A. Unilateral cleft lip (CL)
B. Unilateral cleft lip and palate (CL/P)
C. Bilateral cleft lip and palate (CL/P)
D. Two siblings

21) Cloning is the process of:
• a. synthesis of RNA based on the sequence of DNA.
• b. selective amplification of a specific DNA sequence.
• c. digestion of DNA with restriction endonucleases.
• d. hybridization of a specific DNA sequence with a radio-labeled probe.
• e. visualization of DNA fragments under UV light.

19	20	21
A	D	B

Problem solving

1- Which of the following is NOT required for a PCR reaction?
a. Template DNA
b. Dideoxy-dNTPs (ddNTPs)
c. A thermostable DNA polymerase
d. Primers

2- Which of the following methods is used for separation of DNA fragments?
a. Polymerase chain reaction (PCR)
b. Restriction digestion
c. Gel electrophoresis
d. Bacterial transformation

3- Some bacteria have developed a remarkable resistance to antibiotics. What feature of such bacteria give them the property of antibiotic resistance?
a. They harbor viruses that can attack antibiotic drugs used against bacteria.
b. They contain a plasmid with genes for antibiotic resistance.
c. They have restriction endonucleases protecting them from viral infection.
d. They have a thick cell wall protecting them from the immune system.

1	2	3
B	C	B

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Problem solving

1)

A blood sample from a crime scene was collected. DNA samples of the victim and the potential two suspects were also collected for DNA analysis. The DNA profile is shown. Which one is the criminal?

a. None
b. Suspect 1
c. Both
d. Suspect 2

2)

The polymerase chain reaction for amplification of DNA sequences relies upon the use of:

a. plasmids
b. heat-stable DNA polymerase
c. mRNA
d. restriction enzymes

3)

What is the correct sequence of events in Southern blotting?

a. Transfer of DNA fragments to a membrane followed by separation by electrophoresis and then hybridization with a labelled probe sequence.
b. Separation of DNA fragments by electrophoresis followed by hybridization with a labelled probe sequence and then transfer to a membrane.
c. Separation of DNA fragments by electrophoresis followed by transfer to a membrane and then hybridization with a labelled probe sequence.
d. Hybridization of DNA fragments with a labelled probe sequence followed by separation by electrophoresis and then transfer to a membrane.

19	20	21
D	B	C

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19	20	21
D	B	C

Problem solving

1)

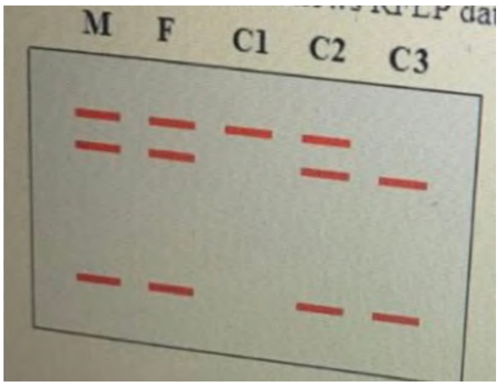
The following figure shows RFLP data done for a family of sickle cell disease.

a. Both parents are carriers and child C3 is normal.

b. Both parents and child C2 are affected.

c. Both parents are carriers and child C3 is affected.

d. Both parents are normal and child C1 is carrier.



2)

The blot shows RFLP mapping for a case of paternity testing.

a. Paternity exclusion

b. Paternity inclusion

19	20	21
A	A	

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Problem solving

1)What is the natural function of restriction enzymes?
a. Protecting bacteria by methylating their own DNA.
b. Protecting bacteria by methylating the DNA of infecting viruses.
c. Protecting bacteria by cleaving their own DNA.
d. Protecting bacteria by cleaving the DNA of infecting viruses.

2)Some bacteria have developed a remarkable resistance to antibiotics. What feature of such bacteria give them the property of antibiotic resistance?
a. They harbor viruses that can attack antibiotic drugs used against bacteria.
b. They contain a plasmid with genes for antibiotic resistance.
c. They have a thick cell wall protecting them from the immune system.
d. They have restriction endonucleases protecting them from viral infection.

19	20	21
D	B	